Unleashing Fourier-Domain Potential: Spatial–Spectral Reconstruction Framework for Remote Sensing Pansharpening

Mengting Ma, Yizhen Jiang, Mengjiao Zhao, Xiaowen Ma, Wei Zhang, and Siyang Song

***Abstract*—Pansharpening aims to generate high-resolution multispectral (HR-MS) images by fusing the corresponding low-resolution multispectral (LR-MS) and high-resolution panchromatic (PAN) images. While the spatial–spectral properties modeling plays a crucial role in generating high-quality HR-MS images, existing approaches suffer from: 1) directly modeling the entangled spatial–spectral properties and 2) lacking task-specific priors for spatial and spectral properties modeling. This article proposes a novel Fourier domain-based approach (Fourier-SSR) to address these problems, where phase and amplitude components of PAN and LR-MS are considered to individually model spatial and spectral properties for the target HR-MS image. Our Fourier-SSR is motivated by the crucial findings in Fourier domain: 1) manifestation of spatial–spectral properties, that is, the spatial and spectral properties of remote sensing images can be individually manifested in their phase and amplitude components in Fourier domain; 2) spatial property-related prior, that is, only reconstructing the phase component of PAN image in Fourier domain, could generate spatial property required for target HR-MS images; and 3) spectral property-related prior, that is, jointly models the amplitude components of PAN and LR-MS images could generate the required spectral property for target HR-MS images. Based on the aforementioned findings, we design a Fourier-guided spatial mixer (FSpa-Mixer) and a Fourier-guided spectral mixer (FSpe-Mixer), which innovatively employ complex feature interaction strategies to individually reconstruct the phase and amplitude components for target HR-MS. Experiments show that our methods unleash Fourier domain potential in individually modeling spatial and spectral properties for the target HR-MS image, leading to superior performance over previous state-of-the-art (SOTA). Our code is provided in the https://github.com/Florina2333.**

***Index Terms*—Complex feature interaction, pansharpening, remote sensing.**

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Mengting Ma is with the School of Computer Science and Technology, Zhejiang University, Hangzhou 310027, China.

Yizhen Jiang, Mengjiao Zhao, and Xiaowen Ma are with the School of Software Technology, Zhejiang University, Hangzhou 310027, China.

Wei Zhang is with the School of Software Technology, Zhejiang University, Hangzhou 310027, China, and also with the Innovation Center of Yangtze River Delta, Zhejiang University, Jiaxing, Zhejiang 314103, China (e-mail: cstzhangwei@zju.edu.cn).

Siyang Song is with HBUG Lab, University of Exeter, EX4 4PY Exeter, U.K. (e-mail: s.song@exeter.ac.uk).

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I. Introduction

With the rapid advancement of satellite sensing techniques, the applications and demand for high-resolution multispectral (HR-MS) images have become essential in various fields such as agriculture monitoring, urban planning, and geological exploration [1]. Due to various physical limitations, a single sensor cannot acquire remote sensing images with rich spatial–spectral properties (i.e., HR-MS images). Instead, existing remote sensing satellites provide only panchromatic (PAN) and low-resolution multispectral (LR-MS) images. There are entangled spatial–spectral properties contained in both LR-MS and PAN images, where LR-MS possesses rich spectral properties but fewer spatial properties, while PAN provides rich spatial details but poor spectral properties [2]. As a result, recent pansharpening approaches have been widely investigated to obtain HR-MS images by reconstructing spatial and spectral properties from the acquired PAN and LR-MS images.

Early pansharpening approaches, including component substitution (CS) algorithm [3], [4], multiresolution analysis (MRA)-based method [5], [6], and variational optimization (VO) [7], were designed based on specific assumptions with manual parameter settings. For example, LRTCP [8] introduces a haze-line prior and designs the more parsimonious VO-based model, that is, a low-tubal-rank tensor completion framework, to reconstruct HR-MS images while preserving spatial–spectral relationships. However, these pansharpening approaches can only model limited spatial–spectral properties from LR-MS and PAN images. Recently, convolutional neural networks (CNNs) have been widely introduced for pansharpening due to their excellent capabilities in modeling task-related spatial–spectral properties from the input PAN and LR-MS images. For example, the VOGNet [9] designs the CNN-based fusion network to generate a prior fusion result, followed by a CNN-based adaptive recovery model to refine spatial details. However, existing CNN-based paradigms [10], [11], [12] directly model the entangled spatial–spectral properties from either the combination of LR-MS and PAN images or each of them, without individually modeling spatial and spectral properties for the target HR-MS image. Alternatively, Transformer-based methods [13], [14] have also been adopted to more effectively model long-range dependency between PAN and LR-MS images, which also jointly reconstruct entangled spatial–spectral properties for the required HR-MS image.

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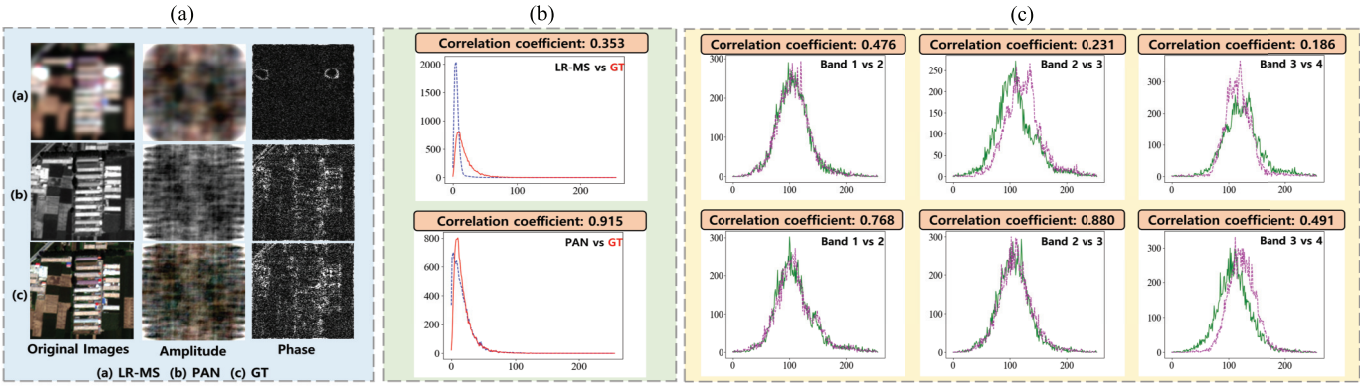


Fig. 1. Motivation. (a) Visualization for the amplitude and phase components of LR-MS, PAN, and GT images. The amplitude retains spectral property (ground object color), while the phase presents spatial structures. (b) Spatial relationship comparison between LR-MS images/PAN images, and corresponding GT images in the phase component. (c) Spectral relationship of LR-MS images (first line) and those of corresponding GT images (second line) in the amplitude component. The correlation coefficient represents the correlation between the two distribution curves; the larger the value, the greater the correlation.

However, modeling both spatial and spectral properties in an entangled manner may lead to mutual interference between them, resulting in HR-MS images obtained suffering from unclear spatial details or distorted spectra (Problem 1). In addition, all of these approaches directly decode the target HR-MS image from the entangled spatial–spectral properties without specifically considering task-specific prior knowledge, that is, the unique spatial/spectral property-related relationships for pansharpening, which limits their generalization capabilities for spatial–spectral properties modeling (Problem 2).

As discussed in [15], [16], and [17], the discrete Fourier transform (DFT) could decouple task-related properties to the corresponding Fourier components in the Fourier domain. Thus, we perform the Fourier components analysis of the pansharpening dataset, as shown in Fig. 1. Specifically, Fig. 1(a) visualizes the Fourier component results of an example provided by the GaoFen-2 dataset, which intuitively presents that the spatial–spectral properties could be individually manifested in the corresponding Fourier components (Finding 1). Fig. 1(b) and (c) shows plots based on the 39095 pairs of PAN/LR-MS/GT images in the GaoFen-2 dataset, based on which two spatial–spectral properties-related priors can be obtained: 1) the phase components of PAN images reflect similar spatial property with those of their target HR-MS images, suggesting that the spatial property of the target HR-MS image could potentially be reconstructed from only the corresponding PAN image's phase (Finding 2); and 2) the amplitude components of LR-MS images reflect different spectral distributions from those of their target HR-MS images, indicating that it is not feasible to reconstruct spectral property of the target HR-MS image only from the LR-MS image (Finding 3).

Inspired by the aforementioned findings, as shown in Fig. 2, we propose the first single Fourier-domain framework for pansharpening (called Fourier-SSR) which: 1) decomposes spatial and spectral properties of the input PAN and LR-MS images into phase and amplitude components in the Fourier domain, to individually model spatial and spectral properties for the target HR-MS image by individually reconstructing phase and amplitude components (addressing Problem 1); and 2) individually combines spatial property-related and spectral property-related relationship priors to achieve phase and amplitude components reconstruction, which promotes the spatial–spectral properties modeling ability (addressing Problem 2). Our approach is inspired by [18], which claims that applying different complex feature interaction strategies between Fourier representations could reconstruct phase and amplitude components (intuitive explanation shown in Section III-C). It includes a novel Fourier-guided spatial mixer (FSpa-Mixer) and a novel Fourier-guided spectral mixer (FSpe-Mixer). Specifically, the FSpa-Mixer applies the explicit feature interaction operation (i.e., Hadamard product) to the PAN-related Fourier representation for reconstructing the phase component of the Fourier representation learned from the interaction operation, which fosters spatial property restoration for the target HR-MS image. Meanwhile, our FSpe-Mixer forms several complex numbers to represent the LR-MS-related and PAN-related Fourier representations, respectively. Then, the channel-wise implicit feature interaction operation is designed by introducing a novel Fourier-MLP for processing these complex numbers, which reconstructs amplitude components of Fourier representations learned from the Fourier-MLP. This module facilitates generating authentic spectral properties. The main contributions of this article are summarized as follows.

1) We propose the first pansharpening approach that individually models spatial and spectral properties of the target HR-MS image from the given PAN and LR-MS images, which is designed based on three crucial pansharpening-related findings.
2) We propose novel FSpa-Mixer and FSpe-Mixer, which innovatively adopt different complex feature interaction strategies to reconstruct phase and amplitude components in the Fourier domain, which facilitates the model's ability to achieve spatial and spectral properties modeling for the target HR-MS image.
3) Extensive experimental results on various benchmarks show that our method not only achieves favorable performance against previous state-of-the-art (SOTA) approaches, but also generalizes well in real-world full-resolution scenes.

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Novelties: The frequency-based pansharpening models (e.g., SFIINet [19], MSDDN [20], and PDDIN [21]) design a Fourier-pixel domain dual-branch network, Fourier domain and pixel domain simultaneous modeling PAN and LR-MS image features. Especially in the Fourier domain, they apply the DFT operation to process PAN/LR-MS image features to obtain their Fourier features and then individually extract amplitude and phase components from the Fourier feature of the LR-MS/PAN image. Finally, they adopt a stacked convolutional network on a combination of component features (i.e., amplitude or phase components) learned from PAN and LR-MS images. Our proposed method differs from them in the following ways.

1) Our whole modeling process only occurs in the Fourier domain, instead of using Fourier domain features as the supplementary cues for spatial-domain analysis.
2) Our model fully combines the spatial–spectral property priors in Fourier components (shown in Fig. 1) rather than modeling all component features of PAN and LR-MS.
3) Our model innovatively exploits a complex feature interaction rule to directly model Fourier features, rather than modeling the component features (i.e., amplitude or phase components) extracted from Fourier features.

II. Related Work

A. Pansharpening Methods

Researchers have been working on pansharpening for nearly four decades, and the methods for this task can be generally classified into four distinct categories [22]: CS-based methods, MRA-based methods, VO-based methods, and deep learning (DL)-based methods.

CS-based methods are alternatively known as spectral methods. They work as follows [3], [4]: first, an LR-MS image is converted into a new space where spatial and spectral information are independent of each other; then, the spatial part of the LR-MS image is replaced with a histogram-matched PAN image; finally, the fusion image is generated by an inverse transformation operation. The fused images obtained by CS-based methods are very rich in spatial detail, but they may suffer from spectral distortion. MRA-based methods rely on extracting and injecting spatial information and are, therefore, also known as spatial methods. To obtain HR-MS images, these methods employ MRA tools [23], [24] to extract high-frequency components from PAN images and subsequently inject these components into interpolated LR-MS images. These methods retain spectral information better than CS-based methods, but may encounter some spatial degradation problems. VO-based methods investigate the interconnections between PAN, LR-MS, and HR-MS images. To generate HR-MS images, they formulate an energy function incorporating diverse regularization terms such as total variation [25], sparse representation [26], and low-rankness [27]. To maintain the accuracy of spatial details and spectral information during the image fusion process, most methods consider the similarity of spatial information between HR-MS and PAN images, as well as the consistency of spectral information between HR-MS and LR-MS images, when designing the energy function [28]. The fused images generated by these methods can retain the spectral information better while maintaining high resolution. However, the computational process of the VO-based methods is relatively complex and time-consuming.

Neural networks have driven significant advancements in computer vision and graphics over the past several decades. However, it was in 2016 that these networks were first utilized to tackle the pansharpening task. The PNN [29] makes the primitive attempt and introduces the CNN into pansharpening, establishing relationships between PAN, LR-MS, and HR-MS images through three convolutional units, leading to impressive improvements compared with traditional methods. Nevertheless, the simplicity of this network architecture results in limited nonlinear capabilities, leading to inadequate exploitation of spatial information. Consequently, numerous CNN-based deep neural networks are subsequently proposed to produce superior-quality fusion outcomes. For example, the PANNet [30] incorporates a residual learning module similar to that used in the ResNet [31]. The MSDCNN [32] adds multiscale modules on residual connections. The SRPPNN [33] addresses the problem by adopting a progressive spectral enhancement network. The DCFNet [34] enhances performance by incorporating features from different scales into the backbone network using multiple parallel branches. Recognizing that the alignment of the same object in LR-MS and PAN images is inconsistent, the SIPSA-Net [10] incorporates a feature alignment module designed to synchronize features extracted from PAN and LR-MS images. Furthermore, some CNN-based models driven by clear physical principles emerge, such as MHNet [35] and GPPNN [36].

The Transformer architecture was first introduced and has since demonstrated outstanding performance in the field of natural language processing. Given the strengths in modeling long-range dependencies, Transformer-based models are introduced to various vision tasks, for example, image classification, semantic segmentation, and object detection. As a groundbreaking work, HyperTransformer [13] applies this model to the pansharpening task. Specifically, it introduces the vanilla cross-modal Transformer [37] to explicitly capture and model the interactions between PAN and LR-MS images. PanFormer [38] utilizes the Transformer framework to improve spatial resolution while maintaining spectral integrity. Additionally, CTINN [14] incorporates the Transformer architecture to capture long-range dependencies, offering features that complement those extracted by the CNN.

However, recent DL-based pansharpening models directly model coupled spatial–spectral properties, thus tending to mutual interference between modalities in the reconstruction process. Different from them, our approach could individually reconstruct spatial and spectral properties, and generate clear spatial structures and authentic spectra.

B. Fourier-Domain Learning

Numerous studies have focused on leveraging frequency-domain information to enhance model performance in various fields, including image denoising [39], image classification [40], image dehazing [41], camouflaged object detection [42],

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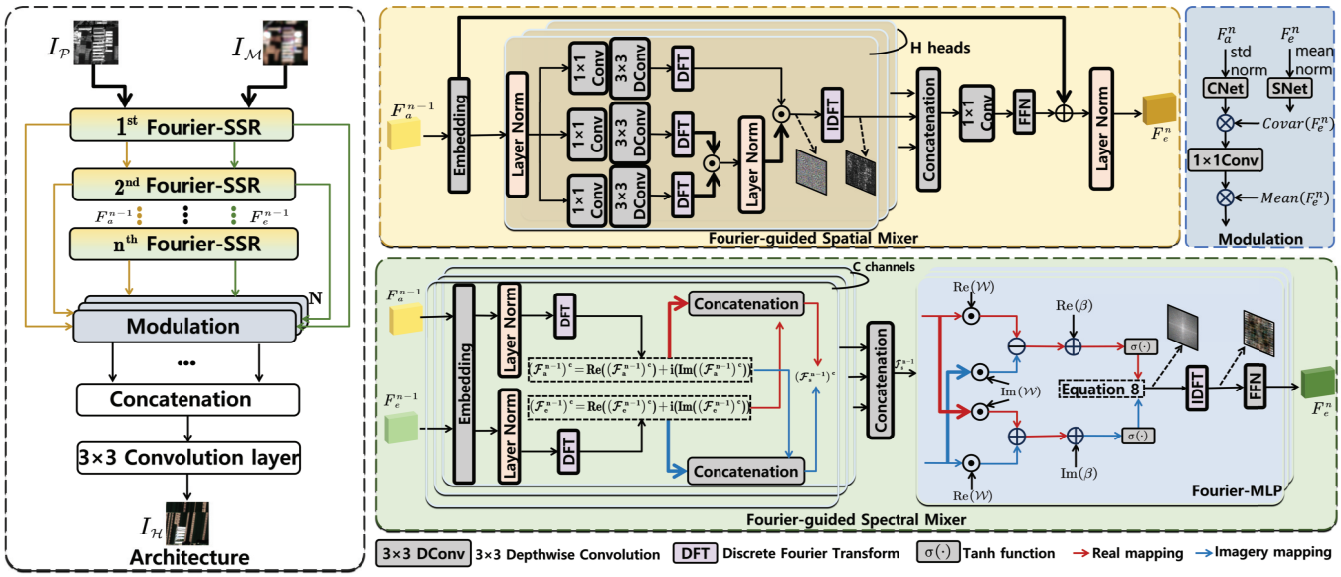


Fig. 2. Overview of the proposed Fourier-SSR paradigm which comprises N pairs of parallel FSpa-Mixer and FSpe-Mixer. Our Fourier-SSR paradigm combines spatial-related and spectral properties-related relationship priors in the Fourier domain and individually models spatial and spectral properties. *cnet* and *snet* both contain three convolutional layers and two ReLU layers in between. The std norm represents normalizing features by the means and standard deviations of channels, while the mean norm normalizes features by the means of its channels.

and more. They typically utilize Fourier transforms to leverage this information. Specifically, in [43], researchers use the Fourier transform to uncover the relationship between haze degradation and the characteristics of amplitude and phase spectra in the frequency domain. In article [44], the Fourier transform is used to extract phase information from images to enhance the model's generalization capability across different domains. Moreover, DBFFT [40] incorporates a Fourier transform branch to learn hidden representations from the frequency domain, thereby enhancing the feature representation capabilities of Vision Transformers. In article [45], the Fourier transform is used to convert hyperspectral images from the spatial–spectral domain to the Fourier domain, to extract and utilize information in the frequency domain.

In recent years, many methods based on frequency-domain learning have also been introduced into the field of pansharpening. For instance, the SFIINet [19] proposes a dual-branch information fusion network that reconstructs spatial–spectral properties from both spatial and Fourier domain perspectives. The MSDDN [20] utilizes the Fourier transform to capture global features and promote the learning of high-frequency information, enabling the capture of multiscale dual-domain information to generate high-quality pansharpened results. Besides, the PDDIN [21] designs the new pyramid dual-domain injection network to inject multiscale details of the PAN image into the LR-MS image in the spatial–frequency domain to aid in the restoration process. However, the aforementioned methods have two key limitations: first, they use all the Fourier features extracted from PAN and LR-MS images without combining specific priors in the Fourier domain, which leads to poor generalization; second, they first extract the Fourier components and then use stacked convolutional layers to encode the Fourier component features. This strategy may destroy the natural correlation between amplitude and phase, resulting in distorted reconstruction results.

III. Methodology

A. Overview

As illustrated in Fig. 2, given an LR-MS image $I_{\mathcal{M}} \in \mathbb{R}^{H\times W\times 4}$ and a PAN image $I_{\mathcal{P}} \in \mathbb{R}^{H\times W\times 1}$, the proposed Fourier-SSR is iterated N times to obtain the final HR-MS image $I_{\mathcal{H}} \in \mathbb{R}^{H\times W\times 4}$. Specifically, each Fourier-SSR includes an FSpa-Mixer that models the spatial property for the required HR-MS image from the PAN image embedding $F_{\mathcal{P}}$, or the PAN-related embedding F_a^{n-1} output from its previous Fourier-SSR (i.e., $(n-1)$th Fourier-SSR) as

$$F_a^n = \text{FSpa} - \text{Mixer}^n\left(F_a^{n-1}\right), \quad F_a^0 = F_{\mathcal{P}} \tag{1}$$

where each FSpa − Mixer applies explicit feature interaction operation to the Fourier representations of *query*, *key*, and *value* generated from F_a^{n-1}, targeting at reconstructing the phase component of the Fourier representation generated by the explicit feature interaction. Meanwhile, each Fourier-SSR also contains an FSpe-Mixer which models the HR-MS image's spectral property from PAN and LR-MS image embeddings (i.e., $F_{\mathcal{P}}$ and $F_{\mathcal{M}}$), or embeddings F_a^{n-1} and F_e^{n-1} output from its previous Fourier-SSR as

$$F_e^n = \text{FSpe} - \text{Mixer}^n\left(F_a^{n-1}, F_e^{n-1}\right) \tag{2}$$

$$F_a^0 = F_{\mathcal{P}}, F_e^0 = F_{\mathcal{M}} \tag{3}$$

where FSpe − Mixer applies channel-wise implicit feature interaction between the Fourier representations of F_a^{n-1} and F_e^{n-1}, aiming at reconstructing the amplitude component of the Fourier representation generated by implicit feature interaction. This way, spatial and spectral properties of the required HR-MS image are independently learned via our FSpa-Mixer and FSpe-Mixer without mutual interference. The features F_e^n and F_a^n output from the nth Fourier-SSR are finally fused and decoded as the final HR-MS image $I_{\mathcal{H}} \in \mathbb{R}^{H\times W\times 4}$. This feature fusion process specifically follows prior works [46] to achieve

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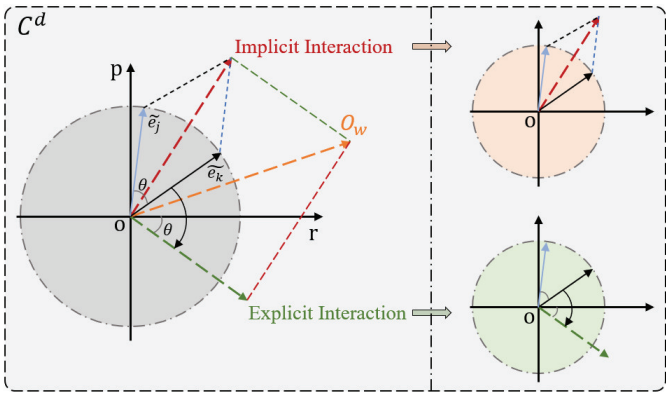


Fig. 3. Visual understanding of the explicit and implicit feature interactions.

spatial–spectral properties aggregation, which first computes the covariance of output feature F_e^n and directly multiplies the covariance with the normalized feature of F_a^n.

B. Training

We choose the simple and standard $\mathcal{L}_1$ loss function to minimize the difference between the predicted HR-MS images H and the ground-truth images G during the network training process as

$$\mathcal{L}_1 = \frac{1}{N}\sum_{i=1}^{N} \|H - G\|_1. \tag{4}$$

C. Intuitive Explanation of Fourier Feature Interaction

In fact, the Fourier domain is considered a complex domain. Thus, the designed FSpe-Mixer exploits implicit feature interaction (i.e., MLP) between Fourier representations generated from PAN and LR-MS images to reconstruct amplitude for the target HR-MS image, and the designed FSpa-Mixer aims to exploit explicit feature interaction (i.e., Hadamard product) between Fourier representations generated from PAN images to reconstruct phase for the target HR-MS image. Why does the implicit feature interaction affect the amplitude, while the explicit feature interaction affects the phase? Inspired by [18], we make the intuitive explanation as follows.

Fig. 3 presents a geometric interpretation of explicit and implicit feature interactions. To have an intuitive understanding, we consider a simple case when the embedding dimension $d = 1$. In complex space, the forms of explicit and implicit interaction could be simplified as

$$g_{\text{ex}}\left(\tilde{\boldsymbol{e}}_j, \tilde{\boldsymbol{e}}_k\right) = \tilde{\boldsymbol{e}}_j^{\alpha_j} \odot \tilde{\boldsymbol{e}}_k^{\alpha_k} \approx \exp\left(i\left(\alpha_j\theta_j + \alpha_k\theta_k\right)\right) \tag{5}$$

$$g_{\text{im}}\left(\tilde{\boldsymbol{e}}_j, \tilde{\boldsymbol{e}}_k\right) = \text{ReLU}\left(W_j\boldsymbol{r}_j + W_k\boldsymbol{r}_k\right) + i\text{ReLU}\left(W_j\boldsymbol{p}_j + W_k\boldsymbol{p}_k\right). \tag{6}$$

As we can see, explicit interaction $g_{\text{ex}}(\cdot)$ mainly affects the phase of features (i.e., θ_j and θ_k), which can be approximately considered as the rotations in the complex vector space, while implicit interaction $g_{\text{im}}(\cdot)$ performs a parallelogram-like transformation in the complex vector space, which mainly affects the modulus instead of the phase (due to the limits in first quadrant). By integrating both implicit and explicit feature interactions, our approach can model the effect in both phase and modulus, thus leading to an improved capacity due to mutual enhancement.

D. Fourier-Guided Spatial Mixer

As illustrated in Fig. 1(a), the spatial property of any remote sensing image could be manifested in its phase component in the Fourier domain. In our work, the FSpa-Mixer aims to model the required spatial property for the target HR-MS image, which is inspired by the evidence that the explicit interaction between Fourier features of the embeddings could facilitate the reconstruction of its phase component (explanation shown in Section III-C). As a result, based on the spatial property-related relationship prior discussed in Fig. 1(b), our FSpa-Mixer applies explicit feature interaction among Fourier representations of the PAN-related *query* embedding $\mathbf{Q}$, *key* embedding $\mathbf{K}$, and *value* embedding $\mathbf{V}$.

As shown in Fig. 2, the FSpa-Mixer starts with separating *query*, *key*, and *value* embeddings obtained by applying the linear projection to the PAN-related embedding F_a^{n-1} as

$$\mathbf{Q} = W_q F_a^{n-1}, \quad \mathbf{K} = W_k F_a^{n-1}, \quad \mathbf{V} = W_v F_a^{n-1} \tag{7}$$

where W_q, W_k, and W_v are linear projections. Then, $\mathbf{Q}, \mathbf{K}$, and $\mathbf{V}$ are split into H heads (see the ablation study of H), where $\mathbf{Q}, \mathbf{K}$, and $\mathbf{V}$ in the hth head are denoted as $\mathbf{Q}^h, \mathbf{K}^h$, and $\mathbf{V}^h$, respectively. Then, each head individually implements Hadamard product (i.e., element-wise multiplication) among its obtained Fourier representations $\mathbf{Q}^h, \mathbf{K}^h$, and $\mathbf{V}^h$ as

$$\mathcal{F}_a^h = \left(\mathcal{F}\left(\mathbf{Q}^h\right) \cdot \overline{\mathcal{F}\left(\mathbf{K}^h\right)}\right) \cdot \mathcal{F}\left(\mathbf{V}^h\right) \tag{8}$$

where $\mathcal{F}$ denotes the DFT and $\overline{\mathcal{F}(\cdot)}$ denotes the conjugate transpose operation. This process reconstructs the phase component $\mathcal{P}(\mathcal{F}_a^h)$ of the Fourier representation $\mathcal{F}_a^h$. Then, we adopt the inverse DFT operation to generate the output embedding F_a^h, reflecting the spatial property that is similar to the required HR-MS image's spatial property as

$$F_a^h = \text{LN}\left(\mathcal{F}^{-1}\left(\mathcal{F}_a^h\right)\right) \tag{9}$$

where LN represents the layer normalization and $\mathcal{F}^{-1}$ denotes inverse DFT operation. Finally, we concatenate each head's output embedding, followed by a feed-forward network (FFN) to generate the feature F_a^n output from the nth FSpa-Mixer.

E. Fourier-Guided Spectral Mixer

As illustrated in Fig. 1(a), the spectral property of any remote sensing image could be manifested in its amplitude component in the Fourier domain. Our FSpe-Mixer aims to model the target HR-MS image's spectral property, which is based on the claims that the implicit interaction between Fourier features of the embeddings can reconstruct its amplitude component (explanation shown in Section III-C). Thus, based on the provided spectral property-related relationship prior shown in Fig. 1(c), our FSpe-Mixer applies the channel-wise implicit feature interaction operation to the Fourier representation of PAN-related embedding and Fourier representation of LR-MS-related embedding, shown in Fig. 2.

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Technically, the FSpe-Mixer takes Fourier-SSR embedding F_e^{n-1} and PAN-related embedding F_a^{n-1} as the inputs, and applies DFT operation to them for obtaining their corresponding Fourier representations $\mathcal{F}_e^{n-1}$ and $\mathcal{F}_a^{n-1}$ as

$$\begin{aligned}\left(\mathcal{F}_e^{n-1}\right)^c &= \mathrm{Re}(\left(\mathcal{F}_e^{n-1}\right)^c) + i(\mathrm{Im}(\left(\mathcal{F}_e^{n-1}\right)^c)) \\ \left(\mathcal{F}_a^{n-1}\right)^c &= \mathrm{Re}(\left(\mathcal{F}_a^{n-1}\right)^c) + i(\mathrm{Im}(\left(\mathcal{F}_a^{n-1}\right)^c))\end{aligned} \tag{10}$$

where $(\mathcal{F}_e^{n-1})^c$ denotes the c_{th} channel of the Fourier representation $\mathcal{F}_e^{n-1}$ and $\mathrm{Re}(\cdot)$ and $\mathrm{Im}(\cdot)$ denote the real and imagery mappings. Subsequently, we concatenate real mappings of Fourier representations $(\mathcal{F}_e^{n-1})^c$ and $(\mathcal{F}_a^{n-1})^c$ as $\mathrm{Re}((\mathcal{F}_s^{n-1})^c)$, as well as their imagery mappings as $\mathrm{Im}((\mathcal{F}_s^{n-1})^c)$. We finally combine the Fourier representations of all channels in complex form as

$$\begin{aligned}\left(\mathcal{F}_s^{n-1}\right)^c &= \mathrm{Re}\left(\left(\mathcal{F}_s^{n-1}\right)^c\right) + i(\mathrm{Im}(\left(\mathcal{F}_s^{n-1}\right)^c)) \\ \mathcal{F}_s^{n-1} &= \left[\left(\mathcal{F}_s^{n-1}\right)^1, \ldots, \left(\mathcal{F}_s^{n-1}\right)^c, \ldots, \left(\mathcal{F}_s^{n-1}\right)^C\right].\end{aligned} \tag{11}$$

This step provides a comprehensive feature space (containing both PAN and LR-MS features) for the subsequent implicit feature interaction. Then, we revisit the MLP mechanism, which can naturally achieve implicit feature interactions [18]. In our work, a novel Fourier-MLP layer is proposed, which combines complex number operation laws with MLP mechanism to foster the Fourier representation learning, to achieve implicit feature interaction between channels of $\mathcal{F}_s^{n-1}$.

Fourier-MLP: Given a complex feature $\mathcal{F}_s^{n-1}$, the designed Fourier-MLP consisting of a complex number weight matrix $\mathcal{W}$ and a complex number bias $\mathcal{B}$ is formulated as

$$\mathcal{Y}^\ell = \sigma\left(\mathcal{Y}^{\ell-1}\mathcal{W}^\ell + \mathcal{B}^\ell\right), \mathcal{Y}^0 = \mathcal{F}_{,s}^{n-1} \tag{12}$$

where $\mathcal{W}^\ell = \mathrm{Re}(\mathcal{W}^\ell) + j(\mathrm{Im}(\mathcal{W}^\ell))$ and $\mathcal{B}^\ell = \mathrm{Re}(\mathcal{B}^\ell) + j(\mathrm{Im}(\mathcal{B}^\ell))$; ℓ denotes the ℓth layer, and σ is an activation function (i.e., tanh). As both $\mathcal{F}_s^{n-1}$ and $\mathcal{W}$ are complex features, we follow the multiplication law of complex numbers. Given two complex numbers $\mathbb{Z}_1 = a + jb$ and $\mathbb{Z}_2 = c + jd$, the multiplication law could be formed as

$$\begin{aligned}\mathbb{Z}_1\mathbb{Z}_2 &= (a + jb)(c + jd) \\ &= ac + j^2bd + jad + jbc = (ac - bd) + j(ad + bc)\end{aligned} \tag{13}$$

thus, we further extend (12) as

$$\begin{aligned}\mathcal{Y}^\ell = \sigma&\left(\mathrm{Re}\left(\mathcal{Y}^{\ell-1}\right)\mathrm{Re}\left(\mathcal{W}^\ell\right) - \mathrm{Im}\left(\mathcal{Y}^{\ell-1}\right)\mathrm{Im}\left(\mathcal{W}^\ell\right) + \mathrm{Re}\left(\mathcal{B}^\ell\right)\right) \\ &+ j\sigma\left(\mathrm{Re}\left(\mathcal{Y}^{\ell-1}\right)\mathrm{Im}\left(\mathcal{W}^\ell\right) + \mathrm{Im}\left(\mathcal{Y}^{\ell-1}\right)\mathrm{Re}\left(\mathcal{W}^\ell\right) + \mathrm{Im}\left(\mathcal{B}^\ell\right)\right).\end{aligned} \tag{14}$$

This way, the channel-wise implicit feature interaction between the Fourier representations $\mathcal{F}_e^{n-1}$ and $\mathcal{F}_a^{n-1}$ is achieved, where $\mathcal{Y}^\ell$ is the Fourier representation output from the implicit feature interaction between channels of $\mathcal{F}_s^{n-1}$. Thus, the amplitude component of the $\mathcal{Y}^\ell$ is reconstructed. Finally, we apply the Inverse DFT operation on $\mathcal{Y}^\ell$ followed by a two-layer FFN, to generate the feature F_e^n (i.e., the output of the nth FSpe-Mixer) containing the modeled spectral property for the target HR-MS image.

F. Modulation Between the FSpa-Mixer and the FSpe-Mixer

The features F_a^n and F_e^n output from the $n_{\mathbf{th}}$ FSpa-Mixer and FSpe-Mixer contain the modeled spatial and spectral properties, respectively. As claimed by [54] that channel-wise statistic of feature reflects its spectral property to some extent, we adopt the statistic of feature F_e^n to modulate the feature F_a^n. As shown in Fig. 2, it makes the adjusted spatial property more statistically consistent with the spectral property, thus ensuring that spatial and spectral properties are mutually compatible when representing the same scene. Inspired by [55], we normalize the feature F_a^n by the means and deviations of its channels to generate a normalized feature $\hat{F}_a^n$, which enhances the stability of subsequent fusion of the spatial–spectral properties. We then normalize the feature F_e^n by means of its channels and calculate its second-order statistic, that is, the covariance matrix $\mathbf{Covar}(F_e^n)$ which models the channel-wise correlations. Finally, we perform a matrix multiplication between $\mathbf{Covar}(F_e^n)$ and normalized feature $\hat{F}_a^n$ and then add channel-wise means of the feature F_e^n (i.e., $\mathbf{Mean}(F_e^n)$) to obtain the feature $F_{\mathcal{H}}$ which contains fused spatial–spectral properties.

IV. Experimental Results

A. Datasets and Benchmark

1) Dataset: We conduct experiments using the widely recognized WorldView-3, QuickBird, and GaoFen-2 datasets.[1] The WorldView-3 dataset consists of instances acquired by the sensor on board the WorldView-3 satellite. This sensor captures data which covers wavelengths from 0.4 to 1 μm, with a spatial resolution of 1.24 m. The QuickBird dataset consists of instances acquired by the sensor on board the QuickBird satellite. This sensor captures data across four spectral bands, covering wavelengths from 0.45 to 0.9 μm, with a spatial resolution of 2.4 m. The images in the GaoFen-2 dataset are collected by the sensor on board the GaoFen-2 satellite, which records data across four spectral bands within the wavelength range of 0.45–0.89 μm. Additionally, this sensor provides a spatial resolution of 3.2 m. The data generation process adheres strictly to Wald's protocol [56], with comprehensive details provided in [57]. As shown in Table III, the WorldView-3 dataset includes 11856 training samples. Additionally, it contains 3619 reduced-resolution and 20 full-resolution testing samples. Each training sample comprises an image triplet in the PAN/LR-MS/GT format, with dimensions of $128\times128\times1$, $32\times32\times4$, and $128\times128\times4$, respectively. The reduced-resolution testing samples include PAN/LR-MS/GT image triplets sized $128\times128\times1$, $32\times32\times4$, and $128\times128\times4$, respectively. Additionally, the full-resolution testing samples consist of image pairs in the PAN/LR-MS format, with sizes of $128\times128\times1$ and $32\times32\times4$, respectively. In the QuickBird dataset, there are 12119 training samples. Additionally, this dataset includes 336 reduced-resolution and 20 full-resolution testing samples. Each training sample contains a PAN/LR-MS/GT image triplet of sizes $128\times128\times1$, $32\times32\times4$ and $128\times128\times4$, respectively. The reduced-resolution testing samples comprise PAN/LR-MS/GT image

[1]Dataset is available at https://github.com/Ritayiyi/Pansharpening-Dataset

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TABLE I

QUANTITATIVE RESULTS OF REDUCED SCALE OVER THREE DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLDFACE, THE SECOND-BEST ONE IS UNDERLINED

	Params	WorldView-3					QuickBird					GaoFen-2				
		PSNR↑	SSIM↑	SAM ↓	ERGAS ↓	SCC↑	PSNR↑	SSIM↑	SAM↓	ERGAS ↓	SCC↑	PSNR↑	SSIM↑	SAM↓	ERGAS ↓	SCC↑
SFIM [47]	—	30.982	0.749	0.076	6.497	0.821	33.463	0.818	0.036	3.071	0.792	37.840	0.917	0.042	3.836	0.806
BROVEY [48]	—	30.857	0.761	0.086	5.506	0.819	30.109	0.837	0.077	2.473	0.739	36.046	0.902	0.036	2.261	0.835
GS [58]	—	35.143	0.807	0.078	5.409	0.825	32.454	0.767	0.033	2.198	0.751	38.328	0.930	0.037	2.826	0.852
SRPPNN [33]	8.96	38.642	0.887	0.066	5.244	0.904	33.966	0.886	0.082	2.840	0.875	40.918	0.961	0.036	2.683	0.911
DCFNet [34]	29.05	34.429	0.862	0.098	6.373	0.568	34.661	0.912	0.072	2.573	0.848	47.055	0.988	0.027	1.270	0.912
CTINN [14]	1.39	39.497	0.889	0.063	5.064	0.962	35.705	0.898	0.038	2.422	0.922	46.875	0.987	0.023	1.302	0.960
ARFNet [49]	1.39	31.263	0.801	0.156	11.224	0.481	32.024	0.843	0.100	4.123	0.668	39.486	0.951	0.059	3.044	0.813
SFIINet [19]	0.85	39.026	0.885	0.064	5.248	0.643	34.459	0.893	0.041	2.811	0.912	47.195	0.988	0.022	1.267	0.980
Hyperformer [13]	439.38	34.121	0.843	0.098	6.356	0.867	34.374	0.869	0.072	2.555	0.914	45.499	0.981	0.034	1.584	0.959
MDCUN [50]	0.98	39.929	0.892	0.056	4.864	0.970	36.070	0.891	0.036	2.377	0.910	48.668	0.991	0.018	1.068	0.986
LGTEUN [51]	2.02	40.124	0.895	0.055	4.791	**0.974**	35.911	0.924	0.045	2.170	0.950	48.283	0.992	0.022	1.126	0.979
MSDDN [20]	0.11	35.127	0.840	0.085	6.232	0.766	34.751	0.872	0.038	2.538	0.935	39.902	0.942	0.039	2.883	0.886
FAFNet [2]	1.22	39.781	0.887	0.056	4.769	0.969	37.209	0.927	0.032	2.212	0.931	49.405	0.984	0.016	0.968	0.977
PDDIN [21]	0.89	38.727	0.872	0.069	5.437	0.934	34.021	0.881	0.047	2.931	0.904	46.242	0.976	0.027	1.278	0.972
FAMENet [52]	1.49	37.219	0.883	0.071	5.737	0.902	37.218	0.929	0.030	**1.996**	**0.953**	42.983	0.981	0.017	1.013	0.978
HFIN [53]	7.94	32.126	0.837	0.083	6.627	0.837	33.616	0.811	0.050	3.174	0.869	41.938	0.966	0.039	2,534	0.894
Fourier-SSR	0.19	**40.623**	**0.905**	**0.054**	**4.741**	0.972	**37.228**	**0.931**	**0.029**	2.022	0.951	**49.541**	**0.994**	**0.015**	**0.955**	**0.990**

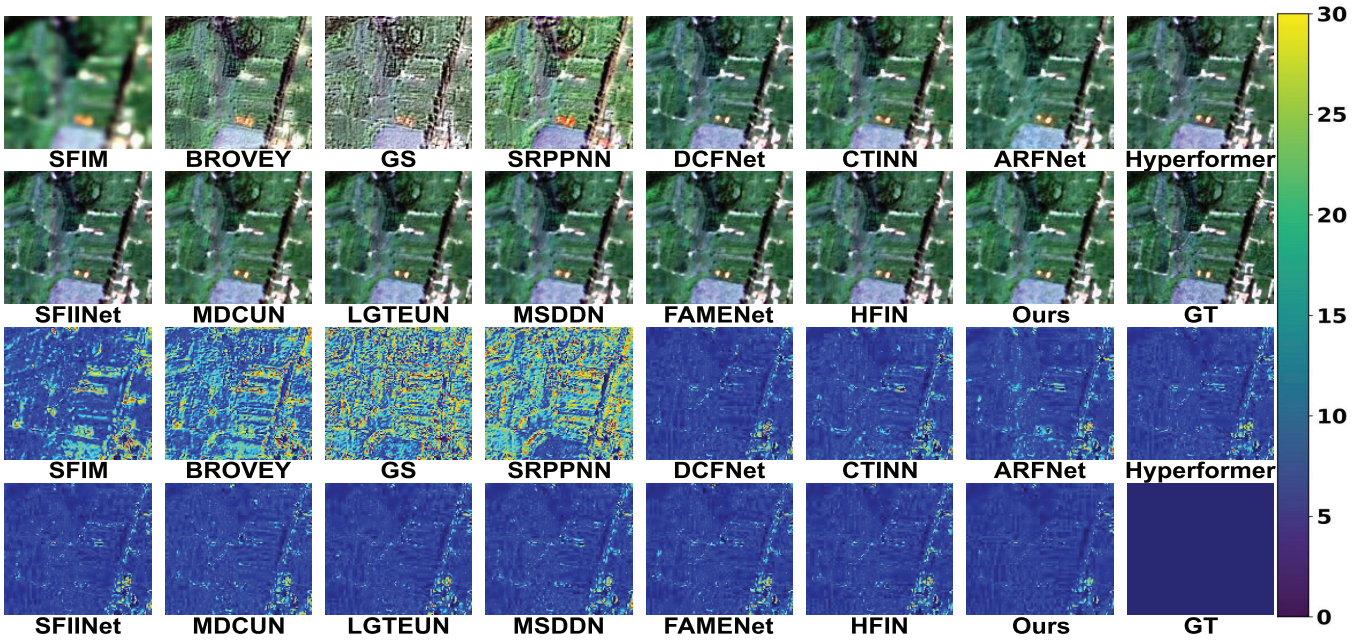


Fig. 4. Qualitative results of reduced-resolution scene on the WorldView-3 dataset. Top group: the fused results. Bottom group: the error between fused results and reference.

triplets sized $128 \times 128 \times 1, 32 \times 32 \times 4$ and $128 \times 128 \times 4$, respectively. Additionally, the full-resolution testing samples consist of PAN/LR-MS image pairs sized $128 \times 128 \times 1$ and $32 \times 32 \times 4$. In the GaoFen-2 dataset, there are 35725 training samples. Additionally, this dataset includes 3350 reduced-resolution and 20 full-resolution testing samples. Each training sample contains a PAN/LR-MS/GT image triplet of sizes $128 \times 128 \times 1, 32 \times 32 \times 4$ and $128 \times 128 \times 4$, respectively. The reduced-resolution testing samples comprise PAN/LR-MS/GT image triplets sized $128 \times 128 \times 1, 32 \times 32 \times 4$ and $128 \times 128 \times 4$, respectively. Additionally, the full-resolution testing samples consist of PAN/LR-MS image pairs sized $128 \times 128 \times 1$ and $32 \times 32 \times 4$.

2) Benchmark: We compare our method with the following two groups of methods.

1) *Learning-Based Methods:* SRPPNN [33], DCFNet [34], CTINN [14], Hyperfomer [13], ARFNet [49], SFIINet [19], MDCUN [50], MSDDN [20], LGTEUN [51], FAFNet [2], PDDIN [21], FAMENet [52], and HFIN [53].
2) *Traditional Methods:* SFIM [47], GS [58], and BROVEY [48]. All comparison methods are retrained on the adopted datasets, without directly using the experimental details in the original articles.

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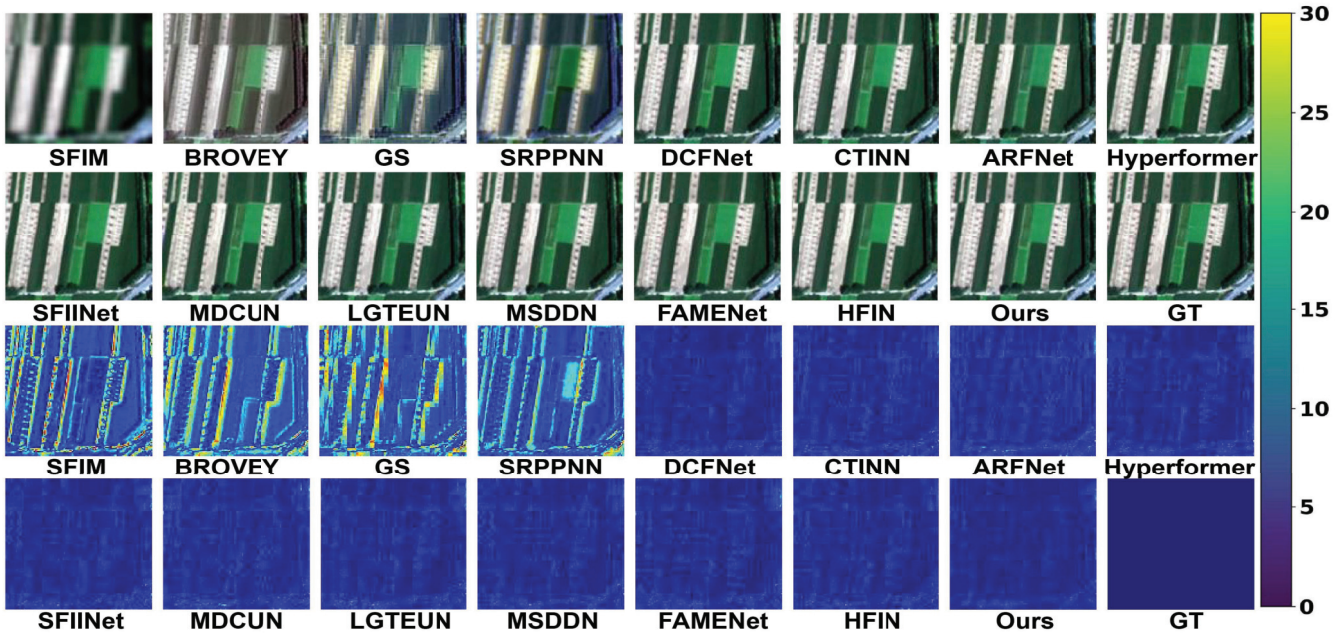


Fig. 5. Qualitative results of reduced-resolution scene on the QuickBird dataset. Top group: the fused results. Bottom group: the error between fused results and reference.

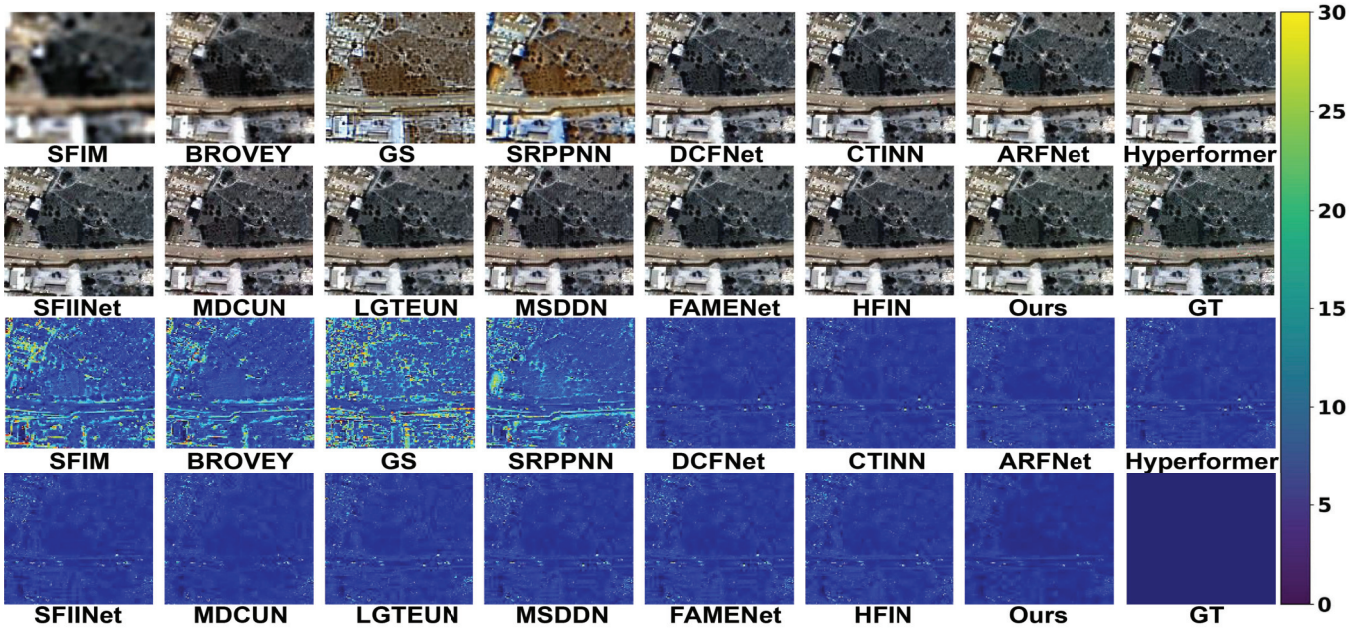


Fig. 6. Qualitative results of reduced-resolution scene on the GaoFen-2 dataset. (Top group) Fused results. (Bottom group) Error between fused results and reference.

B. Evaluation Metrics

Following previous studies on pansharpening, five image quality assessment metrics [59] are employed for evaluation on reduced-resolution images, including spectral angle mapper (SAM), relative dimensionless global error in synthesis (ERGAS), the structural similarity (SSIM), the peak signal-to-noise ratio (PSNR), and the spatial correlation coefficient (SCC) [60]. Specifically, PSNR measures the ratio between peak signal and noise in the image, reflecting the distortion level of the fusion result. SSIM and SCC effectively measure the spatial similarity of the results. SAM and ERGAS are used to measure angular and dynamic range differences between the fused image and the ground-truth image, respectively. Additionally, to further assess the generalization ability of our method, we test it on the corresponding full-resolution real-world scene. Since there are no GT images available for the full-resolution dataset, we use three nonreference metrics to evaluate the performance of the model: spectral distortion index (D_λ), spatial distortion index (D_s), and no-reference quality (QNR) [35]. D_λ concerns spectral shifts between LR-MS and fused HR-MS images, while D_s considers spatial

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TABLE II

EVALUATION ON THE REAL-WORLD FULL-RESOLUTION SCENES FROM THE GAOFEN-2, WORLDVIEW-3, AND QUICKBIRD DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLDFACE, THE SECOND-BEST ONE IS UNDERLINED

Method	GaoFen-2			WorldView-3			QuickBird		
	$D_\lambda\downarrow$	$D_s\downarrow$	QNR↑	$D_\lambda\downarrow$	$D_s\downarrow$	QNR↑	$D_\lambda\downarrow$	$D_s\downarrow$	QNR↑
SFIM [47]	0.082	0.159	0.821	0.137	0.421	0.683	0.219	0.496	0.672
BROVEY [48]	0.138	0.261	0.739	0.136	0.417	0.689	0.226	0.516	0.651
GS [58]	0.074	0.246	0.703	0.141	0.407	0.676	0.231	0.526	0.632
SRPPNN [33]	0.077	0.116	0.817	0.128	0.319	0.757	0.159	0.362	0.700
DCFNet [34]	0.073	0.115	0.839	0.092	0.277	0.774	0.198	0.332	0.723
CTINN [14]	0.072	0.114	0.834	0.072	**0.114**	**0.834**	0.133	0.338	0.732
ARFNet [49]	0.081	0.129	0.817	0.230	0.372	0.709	0.210	0.397	0.652
SFIINet [19]	0.068	0.112	0.847	0.094	0.271	0.706	0.141	0.337	0.732
Hyperformer [13]	0.075	0.119	0.823	0.222	0.357	0.775	0.201	0.420	0.653
MDCUN [50]	0.065	0.110	0.850	0.088	0.257	0.783	0.138	0.339	0.730
LGTEUN [51]	0.063	0.121	0.841	0.091	0.254	0.785	0.137	0.336	0.737
MSDDN [20]	0.069	0.112	0.858	0.149	0.353	0.710	0.149	0.353	0.710
FAFNet [2]	**0.060**	0.108	0.855	0.132	0.331	0.740	0.142	0.341	0.714
PDDIN [21]	0.074	0.121	0.756	0.097	0.261	0.752	0.152	0.381	0.697
FAMENet [52]	0.071	0.153	0.821	0.094	0.259	0.779	0.141	0.339	0.730
HFIN [53]	0.074	0.158	0.814	0.095	0.262	0.771	0.142	0.341	0.723
Fourier-SSR	0.061	**0.107**	**0.860**	**0.085**	0.252	0.792	**0.130**	**0.328**	**0.742**

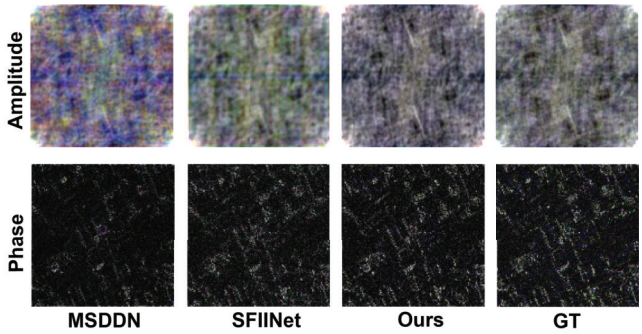


Fig. 7. Phase and amplitude components reconstructed by other Fourier-domain methods and our Fourier-SSR of the GaoFen-2 scene.

disparities between PAN and HR-MS images. QNR is a composite metric measuring spatial and spectral deviations. In these metrics, ideal values for SAM, ERGAS, D_s, and D_λ are 0, while higher values indicate better model performance for the remaining metrics.

C. Training Details

During the training of our networks on the Wordview-3, QuickBird, and GaoFen-2 datasets, the number of iterations is set to 30000, 28000, and 28000, respectively. The Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.999$ is employed for the optimization, and the batch size is set as 4. Our proposed method is trained for 30000 iterations. The initial learning rate is set to 1×10^{-4}, and decays by 0.85 after every 3000 iterations. Additionally, all the experiments are conducted in the PyTorch framework with four NVIDIA RTX A6000 GPUs.

D. Comparisons With SOTA Methods

Evaluation on the Reduced-Resolution Images: In Table I, we report the average evaluation metrics of our proposed method and other selected algorithms on the three datasets, that is, WorldView-3, QuickBird, and GaoFen-2. The best and second-best results are highlighted in boldface and underlined, respectively. It is evident that our method outperforms previous methods in almost all metrics across all three datasets. In particular, all the evaluation metrics of the DL-based methods significantly outperform the traditional methods on the three adopted datasets, which is consistent with the subjective analysis mentioned above and aligns with our understanding. This is because traditional methods are often based on hand-crafted designs, limiting the capability to reconstruct spatial–spectral properties for target HR-MS images. The experimental results also demonstrate that Fourier-SSR excels in spatial details reconstruction and spectral fidelity, as evidenced by its minimum SAM value and maximum SSIM and PSNR values. Furthermore, the fusion effect of the FAMENet and MDCUN is also satisfactory, with relatively good evaluation metrics, roughly ranking second and third, respectively. Moreover, it can be observed from Table I that our Fourier-SSR could achieve the balance between performance and parameters, for example, as compared to the FANENet, our method produces significant PSNR gains (40.623/37.228/49.541 versus 37.219/37.218/42.983 on WorldView-3/QuickBird/GaoFen-2) with fewer parameters (0.19 M versus 1.49 M). In conclusion, the aforementioned analysis indicates that our proposed model can achieve better pansharpening results at a lower complexity.

The subjective evaluation results are shown in Fig. 4 for the WorldView-3 dataset, Fig. 5 for the QuickBird dataset, and Fig. 6 for the GaoFen-2 dataset, using Red-Green-Blue bands composition for presentation. The first two rows present the reconstructed images, while the third and fourth rows display the mean square error (mse) maps between the reconstructed images and the HR-MS images. As evident from the results, our Fourier-SSR is capable of restoring more realistic textures than existing pansharpening methods and markedly enhances the image's information density, including in areas with sparse textures. The traditional methods, that is, SFIM, BROVEY, and GS, struggle to represent details satisfactorily, leading to

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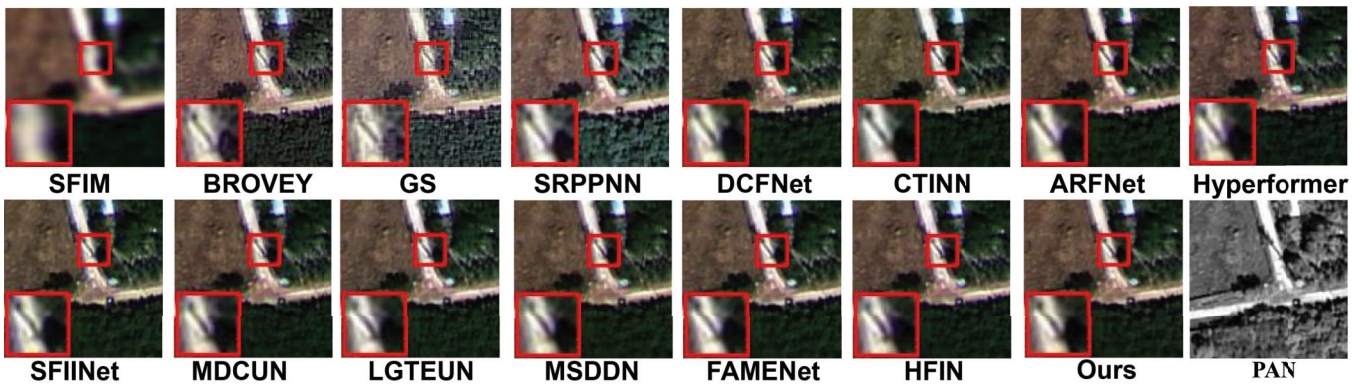


Fig. 8. Visual comparison on the GaoFen-2 full-resolution scene, with some details magnified for better comparison.

blurred building edges and spectral distortion. Similarly, the DL-based methods face challenges in generating sharp and clear textures as well. For instance, SRPPNN and MSDDN produce results with notable color inconsistency. Although MDCUN and FAMENet produce visually more promising results, their error values are still higher than ours. The proposed method generates results that are closer to the original HR-MS images with higher perceptual quality and lower reconstruction errors. These observations further demonstrate the superiority of our method. In addition, the MSDDN and SFIINet are representative Fourier domain pansharpening methods. We obtain the amplitude and phase components reconstructed by these methods, as shown in Fig. 7. This figure shows that the Fourier components reconstructed by our Fourier-SSR are most similar to the reconstructed Fourier components of GT, indicating the effectiveness of our Fourier-domain modeling.

Evaluation on the Full-Resolution Images: To demonstrate the generalization capability of the proposed method, we conduct quantitative evaluations of our pretrained model on 20 unseen full-resolution (real-world) images for each of the GaoFen-2, WorldView-3, and QuickBird datasets. Table II presents the results, and the best and second-best results are highlighted in boldface and underlined, respectively. We could see that the DL-based techniques still perform favorably against their traditional counterparts. In addition, our proposed framework achieves the optimal outcomes for all indices, confirming its superior generalization capability compared to both traditional and DL-based pansharpening methods.

Additionally, we conduct qualitative analysis on the GaoFen-2 dataset. The visual comparison of several methods on a representative full-resolution test scene has been illustrated in Fig. 8. From the figure, it is evident that traditional methods either suffer severe blurring, such as SFIM and GS, or exhibit pronounced spectral distortions, as seen with BROVEY. Even though DL has brought some spatial detail improvements to pansharpening models, modeling spectral properties has been a challenge for these models, exemplified by the ARFNet, HyperTransformer, and MSDDN. Compared to other DL-based methods, our method embeds the spatial–spectral properties priors in the Fourier domain, and exploits the complex feature interaction operation to reconstruct amplitude and phase, finally individually modeling spatial and spectral properties for the target HR-MS image.

Overall, our method demonstrates outstanding performance in both quantitative and qualitative experiments on the full-resolution dataset, indicating its strong generalizability.

TABLE III
DETAILED DATASET INFORMATION (GAOFEN-2, WORLDVIEW-3, AND QUICKBIRD)

Datasets	GaoFen-2	QuickBird	WorldView-3
Raw image	7	5	3
Bit depth	11	10	10
Training set	35725	12119	11856
Testing set	3370	356	3639
LR-MS image size	$32 \times 32 \times 4$	$32 \times 32 \times 4$	$32 \times 32 \times 4$
PAN image size	$128 \times 128 \times 1$	$128 \times 128 \times 1$	$128 \times 128 \times 1$
GT (target HR-MS image) size	$128 \times 128 \times 4$	$128 \times 128 \times 4$	$128 \times 128 \times 4$

E. Ablation Study

To further demonstrate the effectiveness of different settings of our model, in this section, we conduct a range of ablation studies. The experimental results are shown in Tables IV–VII, with the optimal results highlighted in bold.

Effect of Spatial Property Modeling: The Spa-Mixer model the spatial property for the target HR-MS image, which is achieved by adopting explicit feature interaction operation (i.e., multihead dot-product operation) between the query matrix, a key matrix generated by PAN-related Fourier representation. To verify the effectiveness of the Spa-Mixer, we conduct four ablation models over the WorldView-3 and GaoFen-2 datasets, as shown in Tables IV and V. It is observed that the best performance is achieved when selecting $H = 8$ in the Spa-Mixer. If our explicit feature interaction is applied between the Fourier representation of LR-MS and the Fourier representation of PAN (Config IV in Tables IV and V), all indicators significantly reduce, indicating that the introduction of LR-MS images will increase redundancy and excess noise in the process of modeling spatial property.

Effect of Spectral Property Modeling: The Spe-Mixer aims to model the spectral property for the target HR-MS image, which is achieved by applying channel-wise implicit interaction operation (i.e., Fourier-domain MLP) between PAN-related and LR-MS-related Fourier representations. As

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TABLE IV

ABLATION EXPERIMENTS OF THE FSPA-MIXER OVER WORLDVIEW-3 DATASET. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Config	LR-MS feature	PAN feature	H = 8	H = 16	Reduced-resolution Test					Full-resolution Test		
					PSNR↑	SSIM↑	SAM↓	ERGAS↓	SCC↑	D_λ ↓	D_s ↓	QNR↑
I	✗	✓	✗	✓	40.342	0.897	0.059	4.748	0.967	0.089	0.259	0.783
II	✓	✗	✓	✗	39.678	0.865	0.081	4.921	0.931	0.105	0.271	0.751
III	✓	✓	✗	✓	39.921	0.881	0.069	4.755	0.959	0.098	0.265	0.774
IV	✓	✓	✓	✗	40.021	0.891	0.062	4.752	0.962	0.092	0.261	0.779
Ours	✗	✓	✓	✗	**40.623**	**0.905**	**0.054**	**4.741**	**0.972**	**0.085**	**0.252**	**0.792**

TABLE V

ABLATION EXPERIMENTS OF THE FSPA-MIXER OVER GAOFEN-2 DATASET. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Config	LR-MS feature	PAN feature	H = 8	H = 16	Reduced-resolution Test					Full-resolution Test		
					PSNR↑	SSIM↑	SAM↓	ERGAS↓	SCC↑	D_λ ↓	D_s ↓	QNR↑
I	✗	✓	✗	✓	49.312	0.989	0.018	0.959	0.987	0.065	0.109	0.857
II	✓	✗	✓	✗	47.992	0.954	0.024	0.982	0.962	0.083	0.128	0.831
III	✓	✓	✗	✓	48.231	0.977	0.021	0.976	0.972	0.073	0.112	0.845
IV	✓	✓	✓	✗	49.014	982	0.019	0.962	0.981	0.069	0.112	0.853
Ours	✗	✓	✓	✗	**49.541**	**0.994**	**0.015**	**0.955**	**0.990**	**0.061**	**0.107**	**0.860**

TABLE VI

ABLATION EXPERIMENTS OF THE FSPE-MIXER OVER WORLDVIEW-3 DATASET. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Config	PAN feature	LR-MS feature	Common MLP	Fourier-MLP	Reduced-resolution Test					Full-resolution Test		
					PSNR↑	SSIM↑	SAM↓	ERGAS↓	SCC↑	D_λ ↓	D_s ↓	QNR↑
I	✗	✓	✓	✗	39.258	0.891	0.067	4.852	0.962	0.0115	0.267	0.779
II	✓	✗	✓	✗	38.571	0.881	0.072	4.992	0.951	0.109	0.273	0.771
III	✓	✓	✓	✗	40.511	0.901	0.059	4.749	0.968	0.089	0.259	0.787
IV	✗	✓	✗	✓	40.023	0.892	0.062	4.751	0.961	0.092	0.262	0.781
Ours	✓	✓	✗	✓	**40.623**	**0.905**	**0.054**	**4.741**	**0.972**	**0.085**	**0.252**	**0.792**

TABLE VII

ABLATION EXPERIMENTS OF THE FSPE-MIXER OVER THE GAOFEN-2 DATASET. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Config	PAN feature	LR-MS feature	Common MLP	Fourier-MLP	Reduced-resolution Test					Full-resolution Test		
					PSNR↑	SSIM↑	SAM↓	ERGAS↓	SCC↑	D_λ ↓	D_s ↓	QNR↑
I	✗	✓	✓	✗	47.872	0.979	0.024	0.967	0.977	0.078	0.117	0.852
II	✓	✗	✓	✗	46.592	0.972	0.028	0.972	0.971	0.082	0.121	0.845
III	✓	✓	✓	✗	49.322	0.990	0.019	0.959	0.987	0.068	0.109	0.857
IV	✗	✓	✗	✓	48.423	0.982	0.022	0.954	0.982	0.071	0.112	0.853
Ours	✓	✓	✗	✓	**49.541**	**0.994**	**0.015**	**0.955**	**0.990**	**0.061**	**0.107**	**0.860**

shown in Tables VI and VII, we conduct the four ablation models to validate the effectiveness of the Spe-Mixer. Through comparisons, we could obtain that a common MLP cannot effectively model the Fourier domain representation. In addition, implicit feature interaction operation (i.e., Fourier-MLP) is only applied to LR-MS-related Fourier representations, which is suboptimal and cannot produce competitive results.

Effect of the Iteration N: As shown in Fig. 2, the main architecture undergoes N iterations of Fourier-SSR to model the spatial–spectral properties. We conduct five experiments ($N = 1, 2, 3, 4, 5$) to investigate the effect of the iteration N. From the observations in Table VIII, it is evident that the model's performance improves considerably with an increase in the iteration N. However, further increments in iteration

TABLE VIII

ABLATION EXPERIMENTS OF THE ITERATION N OVER THE GAOFEN-2 DATASET

Iteration	PSNR↑	SSIM↑	SAM↓	QNR↑	D_s ↓	D_λ ↓
$N = 1$	47.565	0.967	0.037	0.795	0.128	0.082
$N = 2$	48.381	0.975	0.021	0.839	0.118	0.078
$N = 3$	48.778	0.989	0.017	0.851	0.113	0.068
$N = 4$	**49.541**	**0.994**	**0.015**	**0.860**	**0.107**	**0.061**
$N = 5$	49.359	0.982	0.019	0.843	0.115	0.072

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result in a decreasing trend in three indicators, possibly due to challenges in gradient propagation. Consequently, we employ $N = 4$ as the iteration in all experiments.

V. Conclusion

In this article, we propose Fourier-SSR, an innovative Fourier-domain framework for multispectral pansharpening. To address the two issues of the current modeling framework, that is, modeling the entangled spatial–spectral properties of LR-MS and PAN images, and lacking spatial–spectral properties-related priors, we expand the characteristics of the Fourier domain to the pansharpening and conduct a detailed analysis. Thus, we obtain the three important findings of spatial–spectral properties in the Fourier domain. Based on these findings, we design an FSpa-Mixer and an FSpe-Mixer, which are achieved by adopting the complex feature interaction strategy instead of introducing a convolutional layer. Through reconstructing the amplitude and phase, we could model the spatial and spectral properties, respectively. Additionally, we propose the modulation strategy between modeled spatial and spectral properties for fusing spatial–spectral properties. Finally, we evaluate the performance of Fourier-SSR across three datasets. Our method yields exceptional results, demonstrating the superiority of Fourier-SSR in multispectral pansharpening.

Limitation: Our Fourier-SSR exhibits the major limitation. This Fourier-SSR mainly models the spatial–spectral properties relationships between two input images. However, other fusion tasks may not focus on the spatial–spectral properties, and the recent modeling framework could restrict its applicability in other fusion tasks.

Future Work: In the future, we will delve into the theory of the Fourier domain to promote its further breakthrough innovation in remote sensing image fusion.

References

[1] B. Zhang, D. Wu, L. Zhang, Q. Jiao, and Q. Li, "Application of hyperspectral remote sensing for environment monitoring in mining areas," *Environ. Earth Sci.*, vol. 65, no. 3, pp. 649–658, Feb. 2012.

[2] Y. Xing, Y. Zhang, H. He, X. Zhang, and Y. Zhang, "Pansharpening via frequency-aware fusion network with explicit similarity constraints," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, no. 2, 2023.

[3] B. Aiazzi, S. Baronti, and M. Selva, "Improving component substitution pansharpening through multivariate regression of MS +Pan data," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3230–3239, Oct. 2007.

[4] J. Choi, K. Yu, and Y. Kim, "A new adaptive component-substitution-based satellite image fusion by using partial replacement," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 1, pp. 295–309, Jan. 2011.

[5] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[6] V. Shah, N. H. Younan, and R. L. King, "An efficient pan-sharpening method via a combined adaptive PCA approach and contourlets," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 5, pp. 1323–1335, Apr. 2008.

[7] M. Ghahremani and H. Ghassemian, "A compressed-sensing-based pan-sharpening method for spectral distortion reduction," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 2194–2206, Apr. 2016.

[8] P. Wang et al., "Low-rank tensor completion pansharpening based on haze correction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, no. 3, 2024.

[9] P. Wang, Z. He, B. Huang, M. Dalla Mura, H. Leung, and J. Chanussot, "VOGTNet: Variational optimization-guided two-stage network for multispectral and panchromatic image fusion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 5, pp. 9268–9282, May 2025.

[10] J. Lee, S. Seo, and M. Kim, "SIPSA-net: Shift-invariant pan sharpening with moving object alignment for satellite imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 10161–10169.

[11] W. G. C. Bandara, J. M. J. Valanarasu, and V. M. Patel, "Hyperspectral pansharpening based on improved deep image prior and residual reconstruction," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, no. 7, 2022.

[12] L. He, J. Zhu, J. Li, A. Plaza, J. Chanussot, and B. Li, "HyperPNN: Hyperspectral pansharpening via spectrally predictive convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 8, pp. 3092–3100, Aug. 2019.

[13] W. G. C. Bandara and V. M. Patel, "HyperTransformer: A textural and spectral feature fusion transformer for pansharpening," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 1767–1777.

[14] M. Zhou, J. Huang, Y. Fang, X. Fu, and A. Liu, "Pan-sharpening with customized transformer and invertible neural network," *Proc. AAAI Conf. Artif. Intell.*, vol. 36, no. 3, pp. 3553–3561, Jun. 2022.

[15] C. Li et al., "Embedding Fourier for ultra-high-definition low-light image enhancement," in *Proc. 11th Int. Conf. Learn. Represent.*, Jan. 2023, pp. 12776–12803.

[16] C. Wang, J. Jiang, Z. Zhong, and X. Liu, "Spatial-frequency mutual learning for face super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2023, pp. 22356–22366.

[17] Y. Yang and S. Soatto, "FDA: Fourier domain adaptation for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4085–4095.

[18] Z. Tian, T. Bai, W. X. Zhao, J. Wen, and Z. Cao, "Eulernet: Adaptive feature interaction learning via Euler's formula for CTR prediction," in *Proc. 46th Int. ACM SIGIR Conf. Res. Develop. Inf. Retr.*, 2023, pp. 1376–1385.

[19] M. Zhou et al., "Spatial-frequency domain information integration for pan-sharpening," in *Proc. Eur. Conf. Comput. Vis.*, Jan. 2022, pp. 274–291.

[20] X. He et al., "Multi-scale dual-domain guidance network for pan-sharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5403213.

[21] X. He, K. Yan, R. Li, C. Xie, J. Zhang, and M. Zhou, "Pyramid dual domain injection network for pan-sharpening," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 12862–12871.

[22] X. Meng et al., "A large-scale benchmark data set for evaluating pan-sharpening performance: Overview and implementation," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 1, pp. 18–52, Mar. 2021.

[23] B. Aiazzi, L. Alparone, S. Baronti, and A. Garzelli, "Context-driven fusion of high spatial and spectral resolution images based on oversampled multiresolution analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 10, pp. 2300–2312, Oct. 2002.

[24] L. Alparone, L. Wald, J. Chanussot, C. Thomas, P. Gamba, and L. Mann Bruce, "Comparison of pansharpening algorithms: Outcome of the 2006 GRS-S data-fusion contest," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3012–3021, Oct. 2007.

[25] F. Palsson, J. R. Sveinsson, and M. O. Ulfarsson, "A new pansharpening algorithm based on total variation," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 318–322, Jan. 2014.

[26] M. R. Vicinanza, R. Restaino, G. Vivone, M. Dalla Mura, and J. Chanussot, "A pansharpening method based on the sparse representation of injected details," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 1, pp. 180–184, Jan. 2015.

[27] P. Liu, L. Xiao, and T. Li, "A variational pan-sharpening method based on spatial fractional-order geometry and spectral–spatial low-rank priors," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1788–1802, Mar. 2018.

[28] C. Ballester, V. Caselles, L. Igual, J. Verdera, and B. Rougé, "A variational model for P+XS image fusion," *Int. J. Comput. Vis.*, vol. 69, no. 1, pp. 43–58, Aug. 2006.

[29] G. Masi, D. Cozzolino, L. Verdoliva, and G. Scarpa, "Pansharpening by convolutional neural networks," *Remote Sens.*, vol. 8, no. 7, p. 594, Jul. 2016.

[30] J. Yang, X. Fu, Y. Hu, Y. Huang, X. Ding, and J. Paisley, "PanNet: A deep network architecture for pan-sharpening," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 1753–1761.

[31] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 770–778.

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[32] Q. Yuan, Y. Wei, X. Meng, H. Shen, and L. Zhang, "A multiscale and multidepth convolutional neural network for remote sensing imagery pan-sharpening," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 978–989, Mar. 2018.

[33] J. Cai and B. Huang, "Super-resolution-guided progressive pansharpening based on a deep convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5206–5220, Jun. 2021.

[34] X. Wu, T. Huang, L.-J. Deng, and T.-J. Zhang, "Dynamic cross feature fusion for remote sensing pansharpening," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 14667–14676.

[35] L. Alparone, B. Aiazzi, S. Baronti, A. Garzelli, F. Nencini, and M. Selva, "Multispectral and panchromatic data fusion assessment without reference," *Photogramm. Eng. Remote Sens.*, vol. 74, no. 2, pp. 193–200, Feb. 2008.

[36] S. Xu, J. Zhang, Z. Zhao, K. Sun, J. Liu, and C. Zhang, "Deep gradient projection networks for pan-sharpening," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 1366–1375.

[37] P. Xu, X. Zhu, and D. A. Clifton, "Multimodal learning with transformers: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 12113–12132, Oct. 2023.

[38] H. Zhou, Q. Liu, and Y. Wang, "PanFormer: A transformer based model for pan-sharpening," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2022, pp. 1–6.

[39] X. Fu, J. Huang, X. Ding, Y. Liao, and J. Paisley, "Clearing the skies: A deep network architecture for single-image rain removal," *IEEE Trans. Image Process.*, vol. 26, no. 6, pp. 2944–2956, Jun. 2017.

[40] J. Zeng, L. Huang, X. Bai, and K. Wang, "DBFFT: Adversarial-robust dual-branch frequency domain feature fusion in vision transformers," *Inf. Fusion*, vol. 108, Aug. 2024, Art. no. 102387.

[41] C. Tang and W. Lou, "TCL-net: A lightweight and efficient dehazing network with frequency-domain fusion and multi-angle attention," in *Proc. Asian Conf. Comput. Vis.*, Dec. 2024, pp. 566–582.

[42] Y. Zhong, B. Li, L. Tang, S. Kuang, S. Wu, and S. Ding, "Detecting camouflaged object in frequency domain," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 4504–4513.

[43] H. Yu, N. Zheng, M. Zhou, J. Huang, Z. Xiao, and F. Zhao, "Frequency and spatial dual guidance for image dehazing," in *Proc. Eur. Conf. Comput. Vis.*, Jan. 2022, pp. 181–198.

[44] Q. Xu, R. Zhang, Y. Zhang, Y. Wang, and Q. Tian, "A Fourier-based framework for domain generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14383–14392.

[45] J. Xiao, Y. Liu, S. Zhang, and X. Wei, "Bridging Fourier and spatial–spectral domains for hyperspectral image denoising," in *Proc. 32nd ACM Int. Conf. Multimedia*, Oct. 2024, pp. 8489–8497.

[46] L. Lu, W. Li, X. Tao, J. Lu, and J. Jia, "MASA-SR: Matching acceleration and spatial adaptation for reference-based image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6368–6377.

[47] J. G. Liu, "Smoothing filter-based intensity modulation: A spectral preserve image fusion technique for improving spatial details," *Int. J. Remote Sens.*, vol. 21, no. 18, pp. 3461–3472, Jan. 2000.

[48] A. R. Gillespie, A. B. Kahle, and R. E. Walker, "Color enhancement of highly correlated images. II. Channel ratio and 'chromaticit' transformation techniques," *Remote Sens. Environ.*, vol. 22, no. 3, pp. 343–365, Aug. 1987.

[49] K. Yan et al., "Panchromatic and multispectral image fusion via alternating reverse filtering network," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2022, pp. 21988–22002.

[50] G. Yang, M. Zhou, K. Yan, A. Liu, X. Fu, and F. Wang, "Memory-augmented deep conditional unfolding network for pansharpening," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1778–1787.

[51] M. Li, Y. Liu, T. Xiao, Y. Huang, and G. Yang, "Local-global transformer enhanced unfolding network for pan-sharpening," in *Proc. 32nd Int. Joint Conf. Artif. Intell.*, Aug. 2023, pp. 1071–1079.

[52] X. He, K. Yan, R. Li, C. Xie, J. Zhang, and M. Zhou, "Frequency-adaptive pan-sharpening with mixture of experts," in *Proc. Conf. Innov. Appl. Artif. Intell. (AAAI)*, vol. 38, Mar. 2024, pp. 2121–2129.

[53] J. Tan et al., "Revisiting spatial-frequency information integration from a hierarchical perspective for panchromatic and multi-spectral image fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2024, pp. 25922–25931.

[54] G. Cheng, Z. Li, J. Han, X. Yao, and L. Guo, "Exploring hierarchical convolutional features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6712–6722, Nov. 2018.

[55] Z. Wu, Z. Zhu, J. Du, and X. Bai, "CCPL: Contrastive coherence preserving loss for versatile style transfer," in *Proc. Eur. Conf. Comput. Vis.*, Jan. 2022, pp. 189–206.

[56] L. Wald, T. Ranchin, and M. Mangolini, "Fusion of satellite images of different spatial resolutions: Assessing the quality of resulting images," *Photogramm. Eng. Remote Sens.*, vol. 63, no. 6, pp. 691–699, 1997.

[57] L.-J. Deng et al., "Machine learning in pansharpening: A benchmark, from shallow to deep networks," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 3, pp. 279–315, Sep. 2022.

[58] K. Sandhu, S. S. Patil, M. Pumphrey, and A. Carter, "Multitrait machine{-} and deep-learning models for genomic selection using spectral information in a wheat breeding program," *Plant Genome*, vol. 14, no. 3, Nov. 2021, Art. no. e20119.

[59] G. Yang et al., "PanFlowNet: A flow-based deep network for pan-sharpening," 2023, *arXiv:2305.07774*.

[60] J. Zhou, D. L. Civco, and J. A. Silander, "A wavelet transform method to merge Landsat TM and SPOT panchromatic data," *Int. J. Remote Sens.*, vol. 19, no. 4, pp. 743–757, Jan. 1998.

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